Swati Aggrawal

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PROFESSIONAL SUMMARY

MSc Computer Science graduate from Sapienza Università di Roma specializing in Machine Learning and Artificial Intelligence. Experienced in machine learning experimentation, statistical analysis, model evaluation, and data-driven problem solving using Python, SQL, Pandas, and NumPy. Published IEEE researcher with experience developing analytical pipelines, evaluating model performance, and working with real-world datasets. Interested in risk analytics, quantitative modeling, data science, and data-driven decision making.

EDUCATION

Sapienza Università di Roma **May 2026**

Master of Science in Computer Science

- **Relevant Coursework:** Deep Learning and Applied AI, NLP, Computer Vision, Distributed Systems.
- **Thesis:** Evaluating Pretrained Vision Transformers Under Continual Learning.
- Selected for a competitive international master's program.

Sharda University **July 2023**

Bachelor of Technology - CSE with AI and ML

- **Relevant Coursework:** Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- **Thesis:** Virtual Generative Model Inducing Human Identity to Computers in Metaverse
- Awarded **merit-based scholarship** (Top 5–10% of cohort).

TECHNICAL SKILLS

- **Programming:** Python, SQL, C++, Bash
- **Data Analysis & Statistics:** Pandas, NumPy, Statistical Analysis, Exploratory Data Analysis (EDA), Data Preprocessing, Feature Engineering, Data Visualization
- **Machine Learning:** Classification, Regression, Model Evaluation, Cross Validation, Hyperparameter Tuning, Performance Benchmarking
- **Libraries & Frameworks:** Scikit-Learn, PyTorch, TensorFlow, Keras
- **Deployment & Cloud:** FastAPI, REST APIs, Docker, AWS
- **Tools:** Git, GitHub, Linux

PROJECTS

Lossy Text Compressor [Semantic NLP Pipeline]

- Developed an embedding-based NLP pipeline using GTR-T5 transformer models, vector quantization, and text reconstruction to analyze semantic compression-quality trade-offs.
- Performed systematic experiments across multiple quantization levels and evaluated results using cosine similarity and ROUGE-L metrics.
- Achieved optimal performance at 6-bit quantization with cosine similarity of 0.964 and ROUGE-L score of 0.776.
- Identified non-linear relationships between compression levels and reconstruction quality, providing insights for efficient semantic storage and retrieval.
- Performed data analysis and experimental evaluation using Pandas, NumPy, and Python.
- Conducted quantitative analysis and model evaluation to determine optimal compression-performance trade-offs.

DocDelta [Document Comparison API]

- Developed an NLP-powered document intelligence system using transformer embeddings and semantic similarity techniques for automated comparison of legal and contract documents.
- Designed algorithms to identify added, removed, and modified content while distinguishing numerical changes from textual modifications.
- Built an automated AI workflow that reduced manual document review effort and enabled scalable processing of document revisions.

- Leveraged Sentence Transformers and embedding-based retrieval techniques to analyze semantic similarity between document versions.
- Deployed the application using FastAPI, Docker, AWS ECS Fargate, and CloudWatch for scalable inference and monitoring.
- Evaluated inference performance and optimized deployment architecture, achieving approximately 840ms warm inference latency.
- Experimented with prompt engineering techniques and transformer-based language models to improve semantic analysis workflows.

RESEARCH EXPERIENCE

Continual Learning for Computer Vision (Master's Thesis) — Nov 25 - May 26

- Designed and executed machine learning experiments to evaluate Vision Transformer performance across sequential learning tasks.
- Designed reproducible machine learning experimentation pipelines and comparative evaluation frameworks for transformer-based models.
- Performed statistical analysis and comparative evaluation of model performance across multiple experimental settings.
- Investigated catastrophic forgetting and knowledge retention in transformer-based visual learning systems.
- Automated experiment tracking and performance benchmarking across multiple continual learning scenarios.
- Conducted experiment tracking, performance analysis, and data processing using Pandas and NumPy.
- Documented experimental findings and identified research opportunities for future development of continual learning systems.
- **Technologies :** Python, PyTorch, Vision Transformers, Computer Vision

Machine Learning for Air Pollution [IEEE Publication] — May 2023

- Published an IEEE conference paper reviewing machine learning approaches for air quality prediction.
- Conducted comparative statistical analysis of six machine learning models using multiple performance metrics across environmental datasets.
- Evaluated prediction-based and knowledge-discovery methodologies across six NAAQS pollutants.
- Identified real-time monitoring and IoT integration as important future directions for environmental analytics.